NutriAI - Automated Diet Plan Builder

NutriAI is a full-stack, offline-runnable Streamlit application that generates a 7-day, 3-meal-a-day plan under the 60-second target while enforcing clinical condition filters, allergens, diet modes, diversity, source provenance, and macro/micronutrient analytics.

Item	Implementation
Runtime	Streamlit UI + pure-Python planning package; runs with streamlit run code/app.py
Offline data	10,750 meal candidates plus USDA/RDA/clinical source-reference files in data/; 97/97 template ingredients covered
Clinical rules	IBS, GERD, type 2 diabetes, hypertension, allergens, cross-contamination, diet and cultural constraints
Outputs	7-day plan, explain tables, RDA analytics, benchmarks, downloadable CSV, required persona checks

Architecture

Layer	Responsibility
Data layer	Loads offline food snapshot, USDA ingredient cache, RDA table, clinical lookup CSVs, and clinical rule JSON. No network/API key required durin
Safety layer	Applies hard exclusions before ranking: diet compatibility, cultural constraints, exact allergen tags, cross-contamination risks, FODMAP, GERD, G
Retrieval layer	Embeds meal text and user profile with deterministic dense hashed vectors; scores candidates by cosine similarity.
Ranking layer	Optimizes calorie fit, micronutrient priorities, clinical fit, semantic similarity, and diversity penalties.
Analytics layer	Computes per-meal and per-day nutrients; compares against age/sex RDA/AI targets and flags gaps below 80%.
UI layer	Streamlit dashboard with profile form, plan table, nutrient views, explanations, benchmarks, persona tests, and CSV export.

Data Sources and Rule References

- USDA FoodData Central API guide and ingredient cache: https://fdc.nal.usda.gov/api-guide.html
- NIH Dietary Reference Intakes for RDA/AI targets: https://www.ncbi.nlm.nih.gov/books/NBK56068/
- NHLBI DASH eating plan for sodium cap and potassium/calcium/magnesium/fiber emphasis: https://www.nhlbi.nih.gov/education/dash-eating-plan
- Monash Low FODMAP program for IBS/FODMAP filtering concepts: https://www.monashfodmap.com/
- University of Sydney Glycemic Index database for GI-band filtering concepts: https://glycemicindex.com/

BAX-423 Technique Choices

Technique	Where used	Why it fits
Bloom filter sketching	Allergen and cross-contamination pre-screening in nutriai/bloom.py and nutriai/rules.py	Fast, memory-light membership tests before exact safety checks. This demonstrates s
Dense embeddings	HashedFeature embeddings in nutriai/embedding.py	Transforms noisy meal text and user clinical/diet profile into vectors without needing m
Recommendation ranking	Multi-stage scorer in nutriai/planner.py	Balances calorie target, micronutrient priorities, clinical fit, semantic similarity, and dive

Benchmarks

Metric	Value
Food records scanned	10,750
Safe records for first persona	4,132
Bloom filter bits / hashes	95 bits / 6 hashes
Embedding dimensions	160
Candidate vectors scored	4,132
Total generation time	2.4517 seconds

The implementation intentionally avoids heavyweight infrastructure in the live demo path. The app still demonstrates big-data patterns by scanning a 10,750-row deduplicated offline snapshot, sketching allergen membership, embedding candidate text, and ranking thousands of safe records per profile.

Required Persona Results

Persona	Time	Diversity	Clinical	Allergy	Diet	Nutrition	60 sec
Priya	2.45s	100.0	PASS	PASS	PASS	PASS	PASS
Ravi	3.15s	100.0	PASS	PASS	PASS	PASS	PASS
Mei	2.43s	100.0	PASS	PASS	PASS	PASS	PASS
James	3.25s	100.0	PASS	PASS	PASS	PASS	PASS

Persona-Specific Evidence

- Priya: vegetarian, lactose-free, low-FODMAP plan with no garlic/onion/wheat high-FODMAP meals and no dairy exposure.
- Ravi: gluten-free GERD plan that excludes high-acid tomato/citrus/fried/spicy triggers and pork.
- Mei: vegan, tree-nut-free, low-glycemic plan using fortified foods and low-GI ranking.
- James: pescatarian, soy-free, low-sodium/DASH-oriented plan with potassium, magnesium, and omega-3 priority.

Explainability

Every selected meal includes a ranking explanation with calorie fit, nutrient fit, clinical fit, similarity, and diversity factors. The Sources tab reports USDA/source-reference coverage, API matches, FDC-linked meal rows, and unmapped ingredient counts. Every excluded example lists the exact reason, such as allergen detected, cross-contamination risk, high-FODMAP food for IBS, GERD trigger, high-GI food for diabetes, or high-sodium food for hypertension.

Limitations and Submission Notes

Limitations

- The app does not call APIs at runtime. USDA FoodData Central is incorporated through a committed ingredient-reference cache with 97/97 template-ingredient coverage; live/cached FDC API matches and offline source-reference rows are counted separately.
- Monash Low FODMAP and Glycemic Index data are represented as curated public-guidance lookup mappings, not licensed bulk database exports.
- Allergy and GERD/acidity filtering use internal rule maps matched against meal and USDA ingredient terms because the professor source list does not provide dedicated allergy or GERD bulk datasets.
- Rules are conservative class-project approximations and should be reviewed by a clinician or registered dietitian before real-world use.
- The hashed embedding model is FAISS-ready in spirit but uses numpy cosine retrieval to avoid native binary install risk during demo.
- Meal serving multipliers are used for calorie targeting; production software would use ingredient-level recipe scaling and inventory constraints.

Submission Run Commands

Task	Command
Install	pip install -r code\requirements.txt
Run app	streamlit run code\app.py
Run tests	python -m unittest discover -s code\tests -v

Educational project only. Not medical advice.